

Condorcet's Principle Implies the No Show Paradox

HERVÉ MOULIN*

*Department of Economics,
Virginia Polytechnic Institute and State University,
Blacksburg, Virginia 24061*

Received July 15, 1986; revised June 9, 1987

In elections with variable (and potentially large) electorates, Brams and Fishburn's No Show Paradox arises when a voter is better off not voting than casting a sincere ballot. Scoring methods do not generate the paradox. We show that every Condorcet consistent method (viz., electing the Condorcet winner when there is one) must generate the paradox among four or more candidates. *Journal of Economic Literature* Classification Numbers: 024, 025. © 1988 Academic Press, Inc.

1. INTRODUCTION

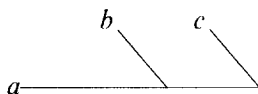
Condorcet's principle says that should a candidate defeat every other candidate in pairwise comparisons (a Condorcet winner), it must be elected. Only pairwise majority comparisons are used to recognize a Condorcet winner and these comparisons are independent of irrelevant alternatives. Thus Condorcet's principle conveys the fundamental idea that the opinion of the majority should prevail, at least when majority comparisons pinpoint an unambiguous winner.

One problem with Condorcet's principle is that it does not tell who should be elected if there is no Condorcet winner. There are many voting rules consistent with Condorcet's principle. Additional properties such as Pareto optimality and positive responsiveness are frequently invoked to choose among those rules (see [3]; also [9]).

We derive another criticism of Condorcet's principle, by comparing elections with variable electorate. This criticism applies to *all* voting rules consistent with Condorcet's principle. We think of a voter facing a choice between participating in the election or not showing up at the polls. Following Brams and Fishburn [1] we call *No Show Paradox* a situation

* Numerous conversations with Bezalel Peleg and Ron Holzman—who suggested one of the examples in the paper—are gratefully acknowledged. I thank Steven Brams and Peyton Young for critical comments. Financial support was provided by the National Science Foundation, under Grant SES 8419465.

where our voter is better off not showing up (as this leads to the election of a candidate whom he prefers). Brams and Fishburn showed that plurality with runoff (a voting rule inconsistent with Condorcet's principle) does generate the No Show Paradox. That the No Show Paradox can also arise in Condorcet consistent methods is demonstrated by the following example (borrowed from [8, p. 122]):



The rule is successive elimination: a against b , next the winner against c . Each vote is taken by simple majority, and possible ties are broken lexicographically. Consider the following seven-voter profile:

	Voter						
	1	2	3	4	5	6	7
Top	c	c	a	a	c	b	b
	b	b	b	b	a	c	c
Bottom	a	a	c	c	b	a	a

In round 1, b beats a (4:3), next b beats c (4:3) in round 2. Suppose that agent 1 does not show up. In the remaining six-voter election, a beats b in round 1 (by the tie breaking rule), next c beats a (4:2). Since agent 1 prefers c to b , he is better off staying home.

Here is another paradoxical feature of our example. Suppose the initial six-voter electorate $\{2, 3, 4, 5, 6, 7\}$ is joined by agent 1. One would expect agent 2 to welcome his "twin" agent, who gives more weight to their common preferences. Yet the addition of agent 1 to the existing electorate results in a net loss to agent 2! We dub this situation the *Twin Paradox*.

It is relatively easy to avoid the No Show and Twin paradoxes: examples of immune voting rules include Borda, plurality, all scoring methods, and more (see Remark 5, Section 4). In the above example of a Condorcet consistent rule subject to the paradoxes, it is conceivable that the particular choice of the tie breaking rule plays a critical role. Question: Is there any Condorcet consistent voting rule immune to the No Show Paradox? To the Twin Paradox?

Apparently a rule immune to the No Show Paradox is immune to the Twin Paradox, too (see Section 3 for details). Thus the second question makes sense only if the answer to the first is negative.

When no more than three candidates are on the floor, we show the existence of Condorcet consistent voting rules immune to both paradoxes:

one such voting rule consists of electing the candidate to whom the smallest majority objects, as suggested by Condorcet himself [2] and later discussed by Fishburn [3] and Kramer [6]. However, when there are four candidates or more, we show that *all* Condorcet consistent voting rules are subject to the No Show Paradox and to the Twin Paradox.

This result parallels an earlier critique of Condorcet's principle by Young [14]. He observed that all Condorcet consistent voting rules violate the *Reinforcement* axiom: this axiom requires that when two disjoint electorates agree on the elected candidate, this candidate still be chosen by the combined electorate. Again Reinforcement compares election outcomes from different electorates, and it is satisfied by all scoring methods. However, Reinforcement and our Participation axiom (which rules out the No Show Paradox; see Section 2) are not logically related: we provide examples of voting rules satisfying one and violating the other (see Remark 5, Section 4). Moreover, Reinforcement—together with anonymity and neutrality—essentially characterizes the scoring methods [15] whereas Participation is compatible with many more (anonymous and neutral) voting rules. In this sense our result is only a critique of Condorcet's principle: that it uses an axiom intuitively weaker than Reinforcement makes the critique altogether stronger.

We state and prove our results for the No Show Paradox in Section 2. In Section 3 we extend them to the Twin Paradox. Section 4 contains some final comments.

2. CONDORCET CONSISTENCY AND THE NO SHOW PARADOX

Given is the finite set A of candidates (outcomes), as well as the set N_∞ of potential voters. The set N_∞ can be finite or countably infinite. Any *finite* subset N of N_∞ is called an electorate. Each voter is endowed with a linear ordering u_i on A : we denote by $L(A)$ the set of all such orderings. Given an electorate N , a profile $u \in L(A)^N$ assigns a linear ordering to each voter $i \in N$.

A *voting rule* is a mapping S associating with every electorate $N \subset N_\infty$ and profile $u \in L(A)^N$, a candidate $S(N, u) \in A$.

The following axiom says that the voting rule S never generates the No Show Paradox:

Participation. For all $N \subset N_\infty$, $|N| \geq 2$, all $u \in L(A)^N$, and $i \in N$, $u_i(S(N/\{i\}, u_{-i})) \leq u_i(S(N, u))$.

In other words a voter never loses by joining the electorate and reporting (sincerely) his preferences.

It is convenient to formulate Condorcet's principle with the following notation:

$$n_{ab}(N, u) = |\{i \in N / u_i(a) > u_i(b)\}| - |\{i \in N / u_i(b) > u_i(a)\}|$$

for all $a, b \in A$, $N \subset N_\infty$, $u \in L(A)^N$.

At a given (N, u) the number n_{ab} is the balance of the number of voters preferring a to b to those preferring b to a . A *Condorcet winner* at (N, u) is a candidate a , necessarily unique, such that $n_{ab} > 0$ for all $b \neq a$. A voting rule is *Condorcet consistent* if it elects the Condorcet winner whenever there is one:

Condorcet Consistency. For all $N \subset N_\infty$, all $u \in L(A)^N$, all $a \in A$, $\{n_{ab} > 0 \text{ for all } b \neq a\} \Rightarrow S(N, u) = a$.

THEOREM. (i) *If A contains three candidates or less, there are voting rules satisfying Participation and Condorcet consistency.*

(ii) *If A contains four candidates or more, and N_∞ contains at least 25 voters, there is no voting rule satisfying Participation and Condorcet consistency.*

The proof of statement (i) is by an example. Given (N, u) we shall denote $m_a = \sup_{b \neq a} n_{ba}$, measuring the size of the largest coalition objecting to a . The Kramer set (or minimax set) is then defined as follows:

$$K(N, u) = \{a \in A / m_a \leq m_b \text{ for all } b \in A\}.$$

Note that a is a Condorcet winner if and only if $m_a < 0$, in which case $m_b > 0$ for all $b \neq a$. Thus whenever a is a Condorcet winner, we have $K(N, u) = \{a\}$, so any single valued selection of K is a Condorcet consistent voting rule.

Suppose now $A = \{a, b, c\}$ contains three candidates and consider the voting rule S electing the first lexicographic element of K (for instance, $m_b = m_c < m_a$ makes b elected). We check that S satisfies Participation.

Suppose it does not: for some N , i , and u we have

$$S(N / \{i\}, u_{-i}) = x, \quad S(N, u) = y, \quad u_i(x) > u_i(y).$$

Distinguish two cases. First assume x is the top candidate of u_i , implying

$$\{n_{zx}(u) = n_{zx}(u_{-i}) - 1 \text{ for all } z \neq x\} \Rightarrow m_x(u) = m_x(u_{-i}) - 1. \quad (1)$$

Similarly

$$\{n_{zy}(u) \geq n_{zy}(u_{-i}) - 1 \text{ for all } z \neq y\} \Rightarrow m_y(u) \geq m_y(u_{-i}) - 1. \quad (2)$$

From $S(N, u) = y$ it follows that $m_y(u) \leq m_x(u)$, and from $S(N/\{i\}, u_{-i}) = x$ it follows that $m_x(u_{-i}) \leq m_y(u_{-i})$. In view of (1) and (2) both inequalities are actually equalities, hence $x, y \in K(N/\{i\}, u_{-i})$ and $x, y \in K(N, u)$ as well. This contradicts the lexicographic tie breaking rule.

The second case is that where x is the middle candidate of u_i and y is the bottom one. Then we have

$$\{n_{zy}(u) = n_{zy}(u_{-i}) + 1 \text{ for all } z \neq y\} \Rightarrow m_y(u) = m_y(u_{-i}) + 1$$

and similarly $m_x(u) \leq m_x(u_{-i}) + 1$. By the same argument as above, we deduce $m_x(u_{-i}) = m_y(u_{-i})$ and $m_x(u) = m_y(u)$, whence the same contradiction.

Since A contains three candidates, these two cases are exhaustive.

In the case where A contains two candidates, our voting rule is simple majority with ties broken in favor of a particular candidate. Apparently it satisfies Participation.

Now to the **proof of statement (ii)**. Fix A , $|A| \geq 4$ and N_∞ , $|N_\infty| \geq 25$. We consider a Condorcet consistent voting rule S satisfying Participation and will derive a contradiction. We prove first an auxiliary result. Given are an electorate $N \subset N_\infty$ and a profile $u \in L(A)^N$. We claim

$$\text{for all } b, a \in A, \quad \{m_b < n_{ba} \text{ and } |N| + m_b + 1 \leq |N_\infty|\} \Rightarrow S(N, u) \neq a. \quad (3)$$

To prove (3) fix b , a satisfying the premises of (3), and yet $S(N, u) = a$. Since S is Condorcet consistent, b cannot be a Condorcet winner, hence $m_b \geq 0$. Observe that given $n = |N|$, all numbers n_{xy} have the same parity as n , and so do all m_x . Therefore $m_b < n_{ba}$ means $n_{ba} - m_b \geq 2$. Setting $p = m_b + 1$ we have

$$m_b < p < n_{ba}; \quad 1 \leq p \quad \text{and} \quad |N| + p \leq |N_\infty|. \quad (4)$$

It follows that we can pick an electorate M , $N \subset M \subset N_\infty$, with cardinality $|N| + p$. In M construct a profile v as follows:

$$\begin{aligned} \text{for } i \in N: \quad & v_i = u_i \\ \text{for } j \in M/N: \quad & v_j \text{ has } a \text{ on top and } b \text{ second.} \end{aligned} \quad (5)$$

Check that b is the Condorcet winner at (M, v) :

$$\begin{aligned} n_{ba}(v) &= n_{ba}(u) - p > 0 \\ n_{bx}(v) &= n_{bx}(u) + p \geq p - m_b(u) > 0 \quad \text{for all } x \neq a, b. \end{aligned}$$

Yet Participation implies that $S(M, v) = a$, as one checks by adding to N the p voters of M/N one at a time. Say $M/N = \{1, \dots, p\}$ and check

$$v_1(S(u, v_1)) \geq v_1(S(u)) = v_1(a) \Rightarrow S(u, v_1) = a$$

$$v_2(S(u, v_1, v_2)) \geq v_2(S(u, v_1)) = v_2(a) \Rightarrow S(u, v_1, v_2) = a$$

and so on... .

This is the desired contradiction establishing the claim (3).

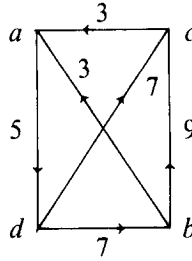
To conclude the proof we construct successively two profiles. The first one has 15 voters. Pick four candidates a, b, c, d in A and a profile u as follows:

Number of voters (15)				
	3	3	5	4
Top	a	a	d	b
	d	d	b	c
	c	b	c	a
	b	c	a	d

(6)

All remaining candidates in $A/\{a, b, c, d\}$ (if any) are ranked below a, b, c , and d .

The weighted majority tournament among the top four candidates (where the weights are the numbers n_{xy}) is as follows:



Invoke (3) successively four times:

$$\{m_d = 5 < n_{db} = 7 \quad \text{and} \quad |N| + m_d + 1 \leq 25\} \Rightarrow S(N, u) \neq b$$

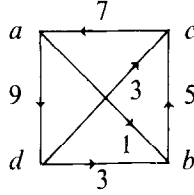
$$\{m_b = 7 < n_{bc} = 9 \quad \text{and} \quad |N| + m_b + 1 \leq 25\} \Rightarrow S(N, u) \neq c$$

$$\{m_a = 3 < n_{ad} = 5 \quad \text{and} \quad |N| + m_a + 1 \leq 25\} \Rightarrow S(N, u) \neq d$$

$$\text{for } x \in A/\{a, b, c, d\}: \quad m_a = 3 < n_{ax} = 15 \Rightarrow S(N, u) \neq x.$$

We conclude $S(N, u) = a$.

Next we augment the electorate N by 4 voters with preferences $c > a > b > d > x$ and denote by (M, v) the new 19-voter profile. The new weighted majority tournament is



Invoking (3) again:

$$\{m_c = 5 < n_{ca} = 7 \quad \text{and} \quad |M| + m_c + 1 \leq 25\} \Rightarrow S(M, v) \neq a$$

$$\{m_b = 3 < n_{bc} = 5 \quad \text{and} \quad |M| + m_b + 1 \leq 25\} \Rightarrow S(M, v) \neq c.$$

On the other hand the repeated application of Participation shows (as above) that $S(M, v)$ must be a or c . This contradiction completes the proof.

3. CONDORCET CONSISTENCY AND THE TWIN PARADOX

The notation is the same as in Section 2. The Twin Paradox for the voting rule S corresponds to the following situation:

- In the electorate $N \subset N_\infty$, with profile u , outcome a is elected:

$$S(N, u) = a.$$

- Agent j with preferences identical to those of agent i , $i \in N$, joins N ($N \cup \{j\} \subset N_\infty$), resulting in a loss to agent i :

$$S(N \cup \{j\}, uu_j) = b, \quad \text{where} \quad u_j = u_i \text{ for some } i \in N \text{ and } u_i(b) < u_i(a).$$

An example of the paradox was given in Section 1 for the successive elimination rule. Here is the axiom ruling out the Twin Paradox:

Twins Welcome. For all $N \subset N_\infty$ and all profiles $u \in L(A)^N$:

$$\{\text{for some } i, j \in N, u_i = u_j\} \Rightarrow \{u_i(S(N/j, u_{-j})) \leq u_i(S(N, u))\}.$$

Evidently Participation implies Twins Welcome (the Twin Paradox implies the No Show Paradox). Conceivably Condorcet's principle may not imply the Twins Paradox as often as the No Show Paradox. The following result dispels any such hope.

COROLLARY TO THE THEOREM. *If A contains four candidates or more, and N_∞ contains at least $25 + |A| \cdot (|A| - 1)/2$ voters, there is no voting rule satisfying Twins Welcome and Condorcet Consistency.*

The proof copies that of statement (ii) in the theorem. Let S be a Condorcet consistent voting rule satisfying Twins Welcome.

Consider an electorate $N_0 \subset N_\infty$ with $|A| \cdot (|A| - 1)/2$ agents and construct a profile $u^0 \in L(A)^{N_0}$ with the following properties:

- For each pair $a, b \in A$ there is exactly one preference u_i , $i \in N_0$, with a on top and b second.
- For each pair $a, b \in A$: $n_{a,b}(N_0, u^0) = 0, +1$, or -1 .

Now pick an electorate N , $N_0 \subseteq N \subset N_\infty$, and a profile $u \in L(A)^N$ such that $u_i = u_i^0$ for $i \in N_0$. Check that Statement (3) still holds: whenever we augment the profile as in (5), we can make sure that each new agent is the twin of some agent in N_0 . Hence the Twins Welcome axiom forces the same argument as Participation did above. The last step of the proof is adapted in the same way. The 15-voter profile (6) is augmented by (N_0, u^0) : in the resulting $(15 + |A| \cdot (|A| - 1)/2)$ profile, the same argument develops.

4. CONCLUDING COMMENTS

Remark 1. The No Show tactic is a particular case of manipulation by sincere truncation of preferences (introduced in [4]). Indeed not showing up is equivalent to complete truncation of one's preferences, namely overall indifference. Fishburn and Brams [4, Theorem 3] noted that Condorcet consistent methods are manipulable by truncation: our theorem can be viewed as a strengthening of theirs. Note that scoring methods like Borda are truncation manipulable, so the truncation of preferences does not bring a specific critique of Condorcet's principle.

Remark 2. The No Show tactic is akin to the familiar free ride concept of the public good literature: the elected candidate is a pure public good (without exclusion) and Participation rules out any incentive to free ride on the decision chosen by the other agents. A similar axiom (dubbed No Free Ride) is studied in [10] in the context of public decision with side-payments (and quasi-linear utilities). It uniquely selects the pivotal mechanism among strategy-proof ones.

Variable population axioms play an increasing role in the social choice literature. In the axiomatic bargaining approach initiated by Nash, Thomson [12, 13] and Lensberg [7] study several conditions relating the choice rules for populations with variable size. Thomson's population

monotonicity axiom (when an additional agent shows up to share the same cake, every other agent tightens his belt) is a kind of dual to Participation.

Remark 3. Our definition of Condorcet consistency uses a strict Condorcet winner. Alternatively we could use a weak Condorcet winner, namely a candidate a (note necessarily unique) such that $n_{an} \geq 0$ for all b different from a . The corresponding consistency axiom is then:

Strong Condorcet Consistency. Given N and u , if there is at least one weak Condorcet winner, then $S(N, u)$ must be one of them.

This axiom is stronger than Condorcet consistency: whenever a strict Condorcet winner exists, there are no other weak Condorcet winners. Therefore our impossibility result (statement (ii) of the theorem) is as strong as can be. On the other hand, the possibility result statement (i)) also holds with Strong Condorcet consistency (by means of the same voting rule).

Remark 4. The simplest generalization of Condorcet's principle consists of requiring a broader majority to defeat the incumbent. Pick a quota q , namely a (real) number $\frac{1}{2} \leq q \leq 1$. Say that candidate a is a q -winner if for no other candidate b we have

$$|\{i \in N/u_i(b) > u_i(a)\}| \geq q \cdot n,$$

where n is the size of the electorate N . The larger the q the more q -winners exist: actually q is strictly greater than $(p-1)/p$ (where p is the size of the issue A) if and only if at least one q -winner exists at every profile. Condorcet's principle generalizes as follows:

q -Consistency. For all $N \subset N_\infty$ and all $u \in L(A)^N$, if there is at least one q -winner at (N, u) then $S(N, u)$ is one of them.

Note that q -Consistency in general does not imply, nor is implied by, q' -Consistency for $q \neq q'$.

Ron Holzman has recently proved the following result ([5]). Suppose A contains four candidates or more and N_∞ is countably infinite. Then Participation and q -Consistency are compatible if and only if q is greater than or equal to $(p-1)/p$.

Remark 5. Comparing Participation and Reinforcement. Given are the set A of candidates and the universal electorate N_∞ . For a voting rule S we consider the following axiom:

Reinforcement. For any two disjoint electorates $N_1, N_2 \subset N_\infty$, any profiles $u_1 \in L(A)^{N_1}$, $u_2 \in L(A)^{N_2}$, and any candidate a ,

$$\{S(N_1, u_1) = S(N_2, u_2) = a\} \Rightarrow S(N_1 \cup N_2, (u_1, u_2)) = a\}.$$

A similar axiom was introduced by Young [14, 15] for social choice functions (multivalued voting rules). With anonymous and neutral social choice functions Young's reinforcement essentially characterizes the scoring methods. With our single valued voting rules, scoring methods yield the simplest examples satisfying reinforcement.

Let p be the cardinality of A and $s_1 \geq s_2 \geq \dots \geq s_p$ be a vector of scores. Given N and u , a candidate x receives s_k points from any voter whose preference gives rank k to x . Call winners at (N, u) those candidates receiving the maximal total score. If there are several winners use a fixed ordering of A to break ties.

This voting rule always satisfies Reinforcement and Participation (actually if all scores s_k are distinct, Participation holds true no matter how we break ties). The obvious proof is omitted.

In the context of preference aggregation, Smith [11] defines a *Separability* axiom which actually implies both Reinforcement and Participation. His axiom considers two disjoint electorates N_1, N_2 with associated profiles u_1, u_2 and two candidates a, b : if the collective preference from (N_1, u_1) prefers a to b or is indifferent and the same holds true for the preferences from (N_2, u_2) then the collective preference from $(N_1 \cup N_2, u_1, u_2)$ must prefer a to b or be indifferent; if in addition either one of the preferences from (N_i, u_i) is strict, then the preference from the combined electorate should also be strict. Smith characterizes scoring aggregation methods on the basis of separability, anonymity, and neutrality.

Thus scoring methods are the main example of voting rules satisfying Participation and Reinforcement. But are there any voting rules satisfying only one of these two axioms? The answer is affirmative.

Here is an example of a voting rule satisfying Reinforcement but not Participation. For a given electorate N denote by $P(N, u)$ the (non-empty) subset of candidates a such that a is ranked last by at most $|N|/|A|$ voters. The elected candidate is chosen in $P(N, u)$ according to a fixed ordering of A .

To check Reinforcement observe that for disjoint electorates N_1, N_2 we have

$$P(N_1 \cup N_2, u_1 u_2) \subset P(N_1, u_1) \cup P(N_2, u_2).$$

To check that Participation does not hold suppose the fixed ordering is lexicographic and consider the six-voter profile:

Top	b	a	a	c	c	a
	a	b	b	b	b	c
Bottom	c	c	c	a	a	b

Here a is elected since $P(N, u) = \{a, b\}$. However, if voter 1 does not show up, b is elected since $P(N/\{1\}, u_{-1}) = \{b\}$.

Now we give an example of a voting rule satisfying Participation but not Reinforcement (the original idea of this example is due to Ron Holzman). For a given electorate N and profile u and for a pair a, b of candidates, denote by $N(a, b)$ the subset of voters who have a on top and b second. The rule elects a candidate a such that for some b , $N(a, b)$ has the largest size among $N(x, y)$, for all $x, y \in A$. Possible ties are broken by a fixed ordering.

To check Participation, observe that upon joining an electorate, a voter can change the election outcome only by making his own top candidate elected. To check that Reinforcement does not hold consider the two following five-voter profiles:

Number of voters			Number of voters		
	3	2		3	2
N_1			N_2		
Top	a	b	Top	a	b
	b	c		c	c
	c	a		b	a

Candidate a is elected by both (N_i, u_i) , $i = 1, 2$. Yet in $(N_1 \cup N_2, u_1 u_2)$, b is chosen.

REFERENCES

1. S. BRAMS AND P. FISHBURN, Paradoxes of preferential voting, *Math. Mag.* **56**, No. 5 (1983), 207–214.
2. A. N. CONDORCET, Marquis de, Essai sur l'application de l'analyse à la probabilité des décisions rendues à la probabilité de voix, Paris, 1785.
3. P. FISHBURN, Condorcet social choice functions. *SIAM J. Appl. Math.* **33**, No. 3 (1977), 469–489.
4. P. FISHBURN AND S. BRAMS, Manipulability of voting by sincere truncation of preferences, *Public Choice* **44** (1984), 397–410.
5. R. HOLZMAN, To vote or not to vote: what is the quota?, mimeo, Rutgers University, New Brunswick, 1987.
6. G. KRAMER, A dynamical model of political equilibrium, *J. Econ. Theory* **16** (1977), 310–334.
7. T. LENSBERG, Stability and collective rationality. *Econometrica* **55** (4) (1987), 935–962.
8. H. MOULIN, Fairness and strategy in Voting in "Fair Allocation" (H. P. Young, Ed.), Amer. Math. Soc., Providence, RI, 1985.
9. H. MOULIN, Choosing from a tournament, *Social Choice and Welfare* **3** (1986), 271–291.

10. H. MOULIN, Characterizations of the pivotal mechanism, *J. Public Econ.* **31** (1986), 53–78.
11. J. SMITH, Aggregation of preferences with variable electorate, *Econometrica* **41**, No. 6 (1973), 1027–1041.
12. W. THOMSON, The fair division of a fixed supply among a growing population, *Math. Oper. Res.* **8** (1983), 319–326.
13. W. THOMSON, Monotonicity, stability and egalitarianism, *Math. Soc. Sci.* **8** (1984), 15–18.
14. H. P. YOUNG, An axiomatization of Borda's rule, *J. Econ. Theory* **9** (1974), 43–52.
15. H. P. YOUNG, Social choice scoring functions, *SIAM J. Appl. Math.* **28** (1975), 824–838.